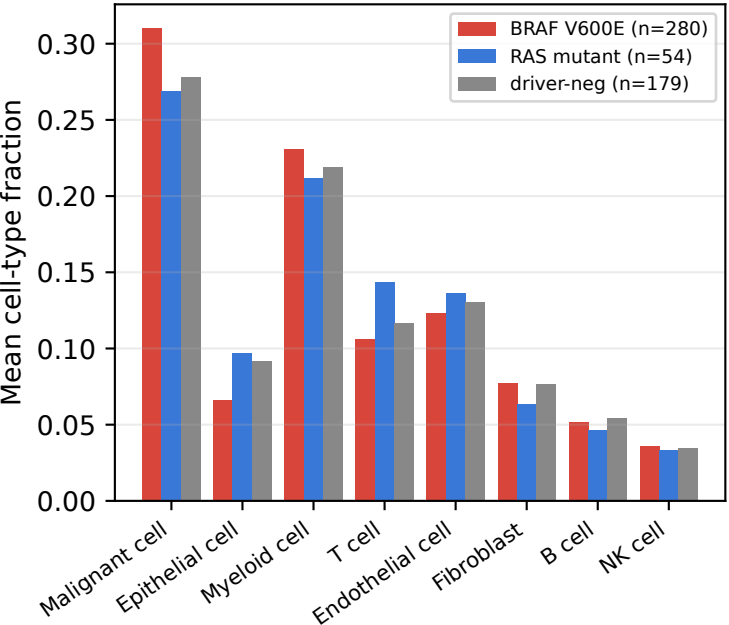
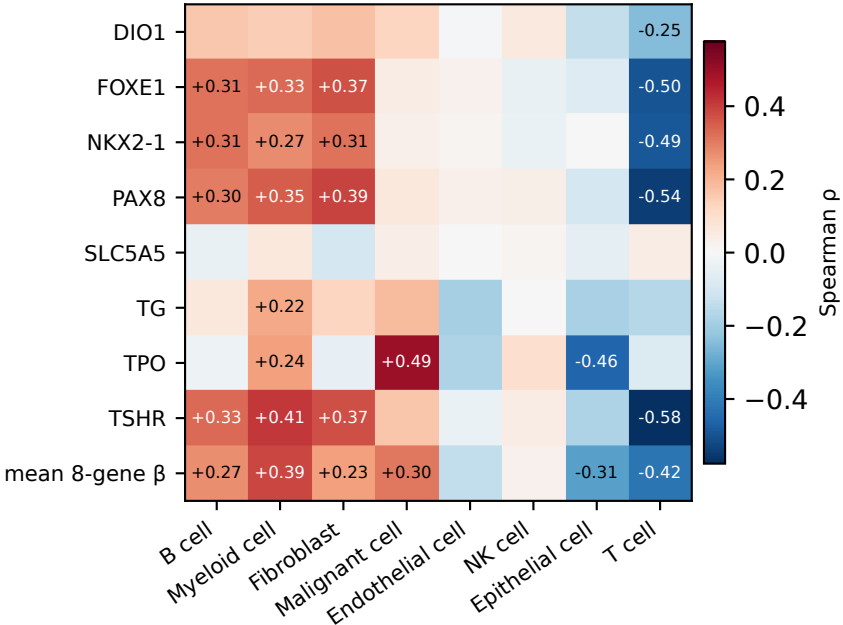


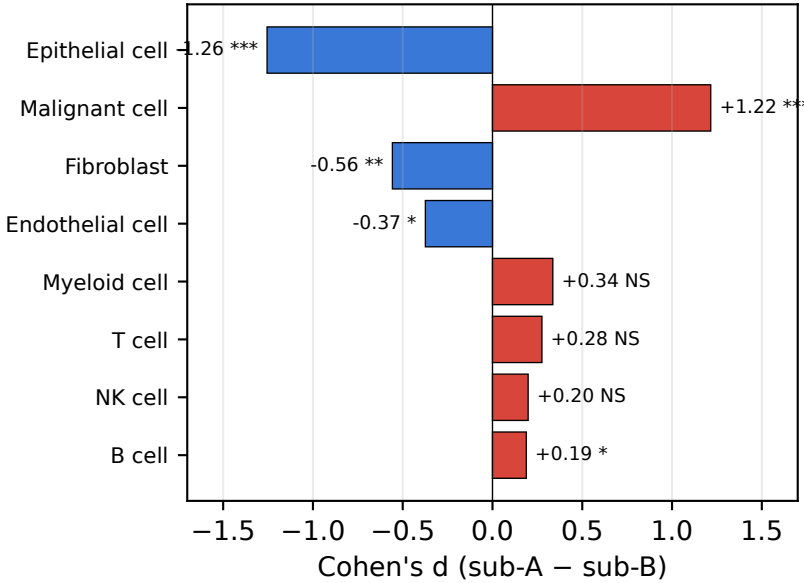
A. Per-driver-class TCGA-THCA composition



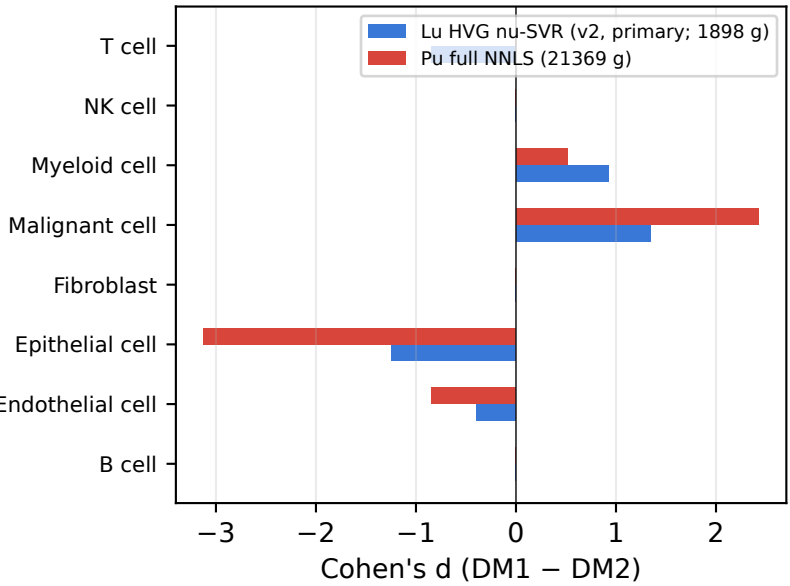
B. 8-gene $\beta \times$ cell-type Spearman ρ



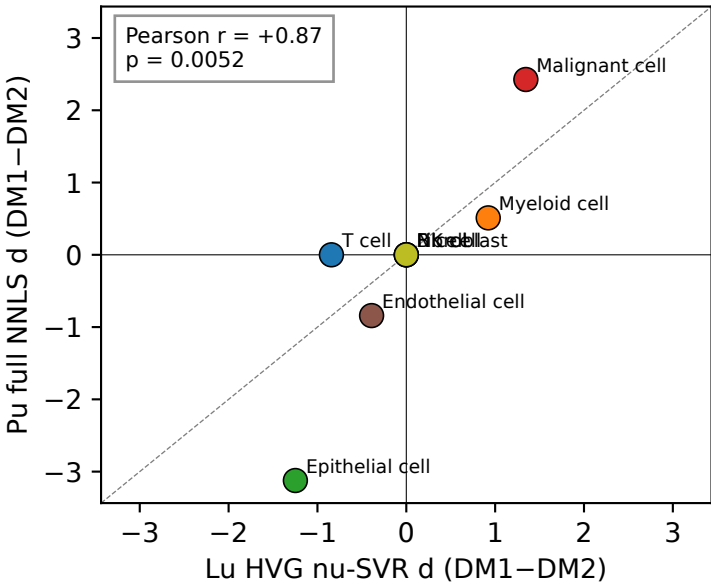
C. DM1 sub-A vs sub-B (Paper 2 boundary)



D. Pu 2021 full-transcriptome ref robustness: Lu HVG vs Pu full



D2. d-vs-d concordance (HVG vs full)



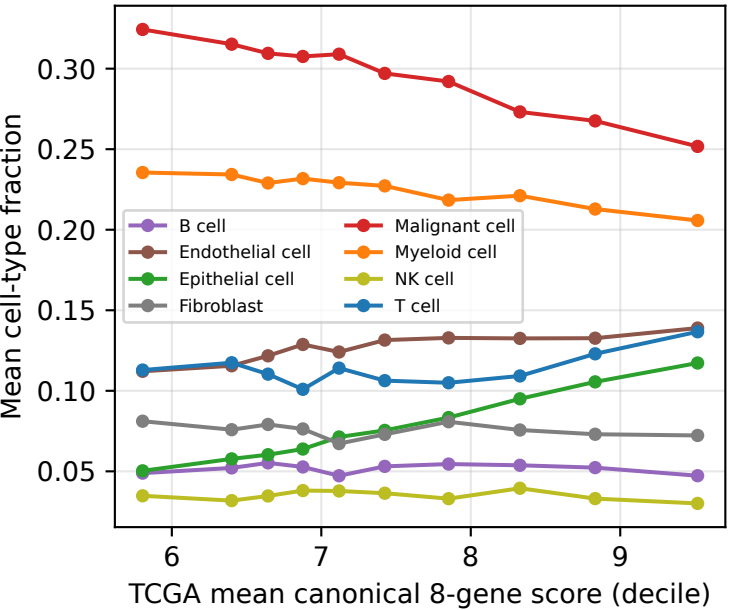
D3. 2-way sign-match summary

Informative cell types ($|d| > 0.05$ in both): 4/8
Sign-match in informative cells: 4/4

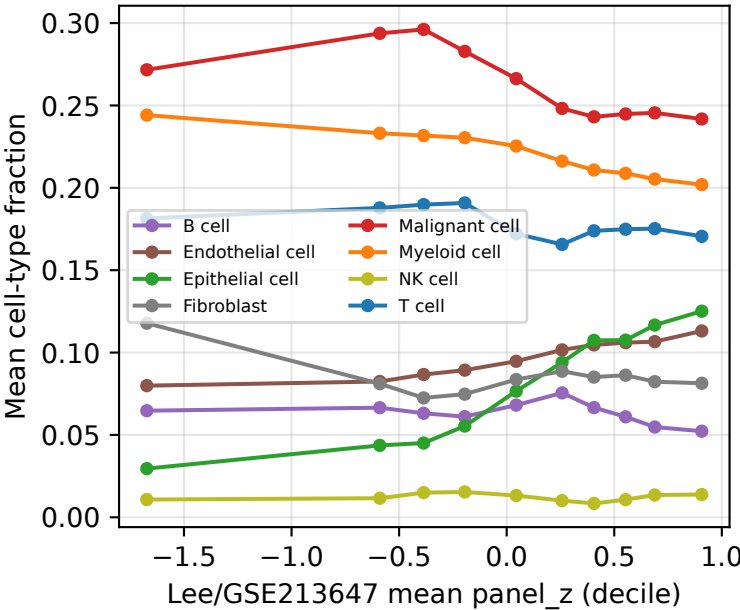
Cell type	Lu HVG	Pu full	match
B cell	+0.00	+0.00	·
Endothelial cell	-0.39	-0.84	✓
Epithelial cell	-1.25	-3.12	✓
Fibroblast	+0.00	+0.00	·
Malignant cell	+1.35	+2.43	✓
Myeloid cell	+0.92	+0.51	✓
NK cell	+0.00	+0.00	·
T cell	-0.84	+0.00	·

· = NNLS sparse-collapse zero (uninformative)
Pu full magnitudes $\geq$ Lu HVG for the 4 high-confidence axes from E pseudotime.

E1. TCGA-THCA pseudotime (score-decile cell-type trajectory)



E2. Lee/GSE213647 pseudotime



E3. Cross-cohort decile-Spearman ρ

Cell type	TCGA ρ	Lee ρ	direction
B cell	+0.02	-0.47	× flipped
Endothelial cell	+0.95	+1.00	✓ DM1↓ DM2↑
Epithelial cell	+1.00	+1.00	✓ DM1↓ DM2↑
Fibroblast	-0.56	+0.08	× flipped
Malignant cell	-0.99	-0.87	✓ DM1↑ DM2↓
Myeloid cell	-0.95	-1.00	✓ DM1↑ DM2↓
NK cell	-0.15	-0.07	✓ DM1↑ DM2↓
T cell	+0.22	-0.60	× flipped

4 perfect-monotonic axes:
Malignant ↓, Epithelial ↑,
Myeloid ↓, Endothelial ↑
(score → DM2; both cohorts $|p| \geq 0.95$)